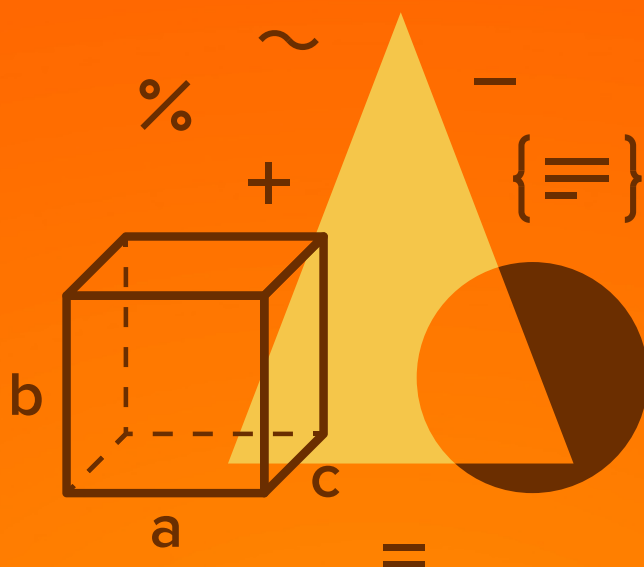




MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Methods of Differentiation

English Medium

EXERCISE (O-1)

1. If $y = \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{dx}$ equals -

(A) $\frac{y}{1-x^2}$ (B) $\frac{y}{x^2-1}$ (C) $\frac{y}{1+x^2}$ (D) $\frac{y}{y^2-1}$

DF0005

2. If $y = \ln \left\{ \frac{x + \sqrt{a^2 + x^2}}{a} \right\}$, then the value of $\frac{dy}{dx}$ is-

(A) $\sqrt{a^2 - x^2}$ (B) $a\sqrt{a^2 + x^2}$ (C) $\frac{1}{\sqrt{a^2 + x^2}}$ (D) $x\sqrt{a^2 + x^2}$

DF0006

3. If $x = y \ln(xy)$, then $\frac{dx}{dy}$ equals-

(A) $\frac{y(x-y)}{x(x+y)}$ (B) $\frac{x(x+y)}{y(x-y)}$ (C) $\frac{y(x+y)}{x(x-y)}$ (D) $\frac{x(x-y)}{y(x+y)}$

DF0007

4. If $(\cos x)^y = (\sin y)^x$, then $\frac{dy}{dx}$ equals -

(A) $\frac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$ (B) $\frac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$ (C) $\frac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$ (D) $\frac{\log \sin y + y \tan x}{\log \cos y - y \cot x}$

DF0008

5. If $y \sec x + \tan x + x^2 y = 0$, then $\frac{dy}{dx} =$

(A) $\frac{2xy + \sec^2 x + y \sec x \tan x}{x^2 + \sec x}$ (B) $-\frac{(2xy + \sec^2 x + \sec x \tan x)}{x^2 + \sec x}$
(C) $-\frac{(2xy + \sec^2 x + y \sec x \tan x)}{x^2 + \sec x}$ (D) None of these

DF0114

6. If $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, then $\frac{dy}{dx}$ is equal to-

(A) 1 (B) 0 (C) -1 (D) -2

DF0015

7. $\frac{d}{dx} \left(\tan^{-1} \left(\frac{\sqrt{x} - x}{1 + x^{3/2}} \right) \right)$ equals- ($x \geq 0$)

(A) $\frac{1}{2\sqrt{x}(1+x)} - \frac{1}{1+x^2}$ (B) $\frac{1}{2\sqrt{x}(1+x)} + \frac{1}{1+x^2}$
(C) $\frac{1}{1+x} - \frac{1}{1+x^2}$ (D) $\frac{1}{1+x} + \frac{1}{1+x^2}$

DF0016

8. If g is the inverse of f and $f'(x) = \frac{1}{1+x^3}$ then $g'(x)$ is equal to-

- (A) $1 + [g(x)]^3$ (B) $\frac{-1}{2(1+x^2)}$ (C) $\frac{1}{2(1+x^2)}$ (D) $\frac{1}{1+[g(x)]^3}$

DF0017

9. If $x^2 + y^2 = 1$, then-

- (A) $yy'' - 2(y')^2 + 1 = 0$ (B) $yy'' + (y')^2 + 1 = 0$
(C) $yy'' + (y')^2 - 1 = 0$ (D) $yy'' + 2(y')^2 + 1 = 0$

DF0018

10. Let f be a function defined for all $x \in \mathbb{R}$. If f is differentiable and $f(x^3) = x^5$ for all $x \in \mathbb{R}$ ($x \neq 0$), then the value of $f'(27)$ is-

- (A) 15 (B) 45 (C) 0 (D) 35

DF0019

11. If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ are given, then -

- (A) $\Delta_1 = 3(\Delta_2)^2$ (B) $\frac{d}{dx} \Delta_1 = 3\Delta_2$ (C) $\frac{d}{dx} \Delta_1 = 3(\Delta_2)^2$ (D) $\Delta_1 = 3(\Delta_2)^{3/2}$

DF0020

12. Let $u(x)$ and $v(x)$ are differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)}\right)' = q$, then

$\frac{p+q}{p-q}$ has the value equal to -

- (A) 1 (B) 0 (C) 7 (D) -7

DF0021

13. Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in A.P., then $f'(a)$, $f'(b)$ and $f'(c)$ are in

- (A) G.P. (B) H.P. (C) A.G.P. (D) A.P.

DF0022

14. Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$, has the value equal to

- (A) 19 (B) 9 (C) 17 (D) 14

DF0023

15. If $f(x) = \begin{cases} x \tan^{-1} x + \sec^{-1}(1/x), & x \in (-1, 1) - \{0\} \\ \pi/2, & \text{if } x = 0 \end{cases}$, then $f'(0)$ is -

- (A) equal to -1 (B) equal to 0 (C) equal to 1 (D) non existent

DF0024

23. If $\frac{d^2x}{dy^2} \left(\frac{dy}{dx} \right)^3 + \frac{d^2y}{dx^2} = K$, then the value of K is equal to

- (A) 1 (B) -1 (C) 2 (D) 0

DF0033

24. Let $f(x) = x + \sin x$. Suppose g denotes the inverse function of f. The value of $g' \left(\frac{\pi}{4} + \frac{1}{\sqrt{2}} \right)$ has the value equal to

- (A) $\sqrt{2} - 1$ (B) $\frac{\sqrt{2} + 1}{\sqrt{2}}$ (C) $2 - \sqrt{2}$ (D) $\sqrt{2} + 1$

DF0036

25. If $x \log_e(\log_e x) - x^2 + y^2 = 4$ ($y > 0$), then dy/dx at $x = e$ is equal to :

- (1) $\frac{e}{\sqrt{4 + e^2}}$ (2) $\frac{(1 + 2e)}{2\sqrt{4 + e^2}}$ (3) $\frac{(2e - 1)}{2\sqrt{4 + e^2}}$ (4) $\frac{(1 + 2e)}{\sqrt{4 + e^2}}$

DF0115

26. If $2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2$, $x \in \left(0, \frac{\pi}{2} \right)$, then $\frac{dy}{dx}$ is equal to :

- (1) $2x - \frac{\pi}{3}$ (2) $\frac{\pi}{3} - x$ (3) $\frac{\pi}{6} - x$ (4) $x - \frac{\pi}{6}$

DF0116

27. Let $f(x) = \log_e(\sin x)$, ($0 < x < \pi$) and $g(x) = \sin^{-1}(e^{-x})$, ($x \geq 0$). If α is a positive real number such that $a = (f \circ g)'(\alpha)$ and $b = (f \circ g)(\alpha)$, then :

- (1) $a\alpha^2 - b\alpha - a = 0$ (2) $a\alpha^2 + b\alpha - a = -2\alpha^2$
(3) $a\alpha^2 + b\alpha + a = 0$ (4) $a\alpha^2 - b\alpha - a = 1$

DF0117

28. If $f(1) = 1$, $f'(1) = 3$, then the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$ is :

- (1) 12 (2) 33 (3) 9 (4) 15

DF0105

29. If $f(4) = g(4) = 2$; $f'(4) = 9$; $g'(4) = 6$ then $\lim_{x \rightarrow 4} \frac{\sqrt{f(x)} - \sqrt{g(x)}}{\sqrt{x} - 2}$ is equal to

- (A) $3\sqrt{2}$ (B) $\frac{3}{\sqrt{2}}$ (C) 0 (D) none of these

DF0118

30. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x) + 2)]^2$. Then $g'(0) :-$

- (1) 4 (2) -4 (3) 0 (4) -2

DF0097

EXERCISE (O-2)

Multiple objective type :

1. If $f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f , then

(A) $g'(2\pi) = \frac{7}{3}$ (B) $g'(2\pi) = \frac{3}{7}$ (C) $g''(2\pi) = \frac{7}{3}$ (D) $g''(2\pi) = 0$

DF0037

2. If $f(x) = x \cdot |x|$, then its derivative is :

(A) $2x$ (B) $-2x$ (C) $2|x|$ (D) $2x \operatorname{sgn} x$

DF0038

3. If $y = \tan x \tan 2x \tan 3x$, ($\sin 12x \neq 0$) then $\frac{dy}{dx}$ has the value equal to

(A) $3 \sec^2 3x \tan x \tan 2x + \sec^2 x \tan 2x \tan 3x + 2 \sec^2 2x \tan 3x \tan x$
 (B) $2y (\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x)$
 (C) $3 \sec^2 3x - 2 \sec^2 2x - \sec^2 x$
 (D) $\sec^2 x + 2 \sec^2 2x + 3 \sec^2 3x$

DF0039

4. Which of the following statements are true ?

(A) If $xe^{xy} = y + \sin^2 x$, then at $x = 0$, $(dy/dx) = 1$.
 (B) If $f(x) \equiv a_0 x^{2m+1} + a_1 x^{2m} + a_2 x^{2m-1} + \dots + a_{2m+1} = 0$ ($a_0 \neq 0$) is a polynomial equation with rational coefficients then the equation $f'(x) = 0$ must have a real root. ($m \in \mathbb{N}$).
 (C) If $(x - r)$ is a factor of the polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0$ repeated m times where $1 \leq m \leq n$ then r is a root of the equation $f'(x) = 0$ repeated $(m-1)$ times.
 (D) If $y = \sin^{-1}(\cos \sin^{-1} x) + \cos^{-1}(\sin \cos^{-1} x)$ then $\frac{dy}{dx}$ is dependent on x .

DF0040

5. Let $f(x) = \frac{\sqrt{x-2}\sqrt{x-1}}{\sqrt{x-1}-1} \cdot x$, then

(A) $f'(10) = 1$ (B) $f'(3/2) = -1$
 (C) domain of $f(x)$ is $x \geq 1$ (D) none

DF0041

6. $\lim_{x \rightarrow \pi} \frac{x^\pi - \pi^x}{x^x - \pi^\pi}$ is equal to -

(A) $\log_{\pi e} \left(\frac{e}{\pi} \right)$ (B) $\log_{\pi e} \left(\frac{\pi}{e} \right)$
 (C) $\tan(\cot^{-1}(\ell n \pi) - \cot^{-1}(1))$ (D) $\tan(\tan^{-1}(1) - \tan^{-1}(\ell n \pi))$

DF0042

7. Let $P(x)$ be the polynomial $x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$. If $P(-3) = P(2) = 0$ and $P'(-3) < 0$, which of the following is a possible value of 'c' ?
 (A) -27 (B) -18 (C) -6 (D) -3

DF0043

8. Two functions f & g have first & second derivatives at $x = 0$ & satisfy the relations,

$$f(0) = \frac{2}{g(0)}, f'(0) = 2g'(0) = 4g(0), g''(0) = 5 f''(0) = 6f(0) = 3 \text{ then -}$$

- (A) if $h(x) = \frac{f(x)}{g(x)}$ then $h'(0) = \frac{15}{4}$
 (B) if $k(x) = f(x) \cdot g(x) \sin x$ then $k'(0) = 2$
 (C) $\lim_{x \rightarrow 0} \frac{g'(x)}{f'(x)} = \frac{1}{2}$
 (D) none

DF0044

9. If $y = x^{(\ln x)^{\ln(\ln x)}}$, then $\frac{dy}{dx}$ is equal to :

- (A) $\frac{y}{x} (\ln x^{\ln x - 1} + 2 \ln x \ln(\ln x))$ (B) $\frac{y}{x} (\ln x)^{\ln(\ln x)} (2 \ln(\ln x) + 1)$
 (C) $\frac{y}{x \ln x} ((\ln x)^2 + 2 \ln(\ln x))$ (D) $\frac{y \ln y}{x \ln x} (2 \ln(\ln x) + 1)$

DF0045

10. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to

- (A) $-\frac{2^y}{2^x}$ (B) $\frac{1}{1 - 2^x}$ (C) $1 - 2^y$ (D) $\frac{2^x (1 - 2^y)}{2^y (2^x - 1)}$

DF0119

11. The differential coefficient of $\sin^{-1} \frac{t}{\sqrt{1+t^2}}$ w.r.t. $\cos^{-1} \frac{1}{\sqrt{1+t^2}}$ is

- (A) $1 \forall t > 0$ (B) $-1 \forall t < 0$ (C) $1 \forall t \in \mathbb{R}$ (D) none of these

DF0120

12. If $f_n(x) = e^{f_{n-1}(x)}$ for all $n \in \mathbb{N}$ and $f_0(x) = x$, then $\frac{d}{dx} \{f_n(x)\}$ is equal to

- (A) $f_n(x) \cdot \frac{d}{dx} \{f_{n-1}(x)\}$ (B) $f_n(x) \cdot f_{n-1}(x)$
 (C) $f_n(x) \cdot f_{n-1}(x) \dots f_2(x) \cdot f_1(x)$ (D) none of these

DF0121

13. If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$, then

- (A) $f(-2) = 0$ (B) $f'(-1/2) = 0$ (C) $f'(-1) = -2$ (D) $f''(0) = 4$

DF0122

14. If $f(\theta) = \tan\left(\sin^{-1}\sqrt{\frac{2}{3+\cos 2\theta}}\right)$, then-

(A) $f'\left(\frac{\pi}{4}\right) = \sqrt{2}$

(B) $\frac{d(f(\theta))}{d(\cos \theta)} = -2$ at $\theta = \frac{\pi}{4}$

(C) $\frac{d(f(\theta))}{d(\cos \theta)} = -\frac{1}{2}$ at $\theta = \frac{\pi}{4}$

(D) $\frac{d(f(\theta))}{d(\sec \theta)} = 1$ at $\theta = \frac{\pi}{3}$

DF0123

15. The functions $u = e^x \sin x$; $v = e^x \cos x$ satisfy the equation

(A) $v \frac{du}{dx} - u \frac{dv}{dx} = u^2 + v^2$

(B) $\frac{d^2 u}{dx^2} = 2v$

(C) $\frac{d^2 v}{dx^2} = -2u$

(D) $\frac{du}{dx} + \frac{dv}{dx} = 2v$

DF0124

EXERCISE (O-3)

Paragraph for Question 1 to 3

Let $a_1 > a_2 > a_3 \dots \dots \dots a_n > 1$; $p_1 > p_2 > p_3 \dots \dots \dots > p_n > 0$; such that $p_1 + p_2 + p_3 + \dots \dots + p_n = 1$. Also

$F(x) = (p_1 a_1^x + p_2 a_2^x + \dots \dots \dots + p_n a_n^x)^{1/x}$ then

1. $\lim_{x \rightarrow 0^+} F(x)$ is

- (A) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n}$ (B) a_1 (C) a_n (D) does not exist

2. $\lim_{x \rightarrow \infty} F(x)$

- (A) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n}$ (B) a_1 (C) a_n (D) does not exist

3. $\lim_{x \rightarrow -\infty} F(x)$

- (A) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n}$ (B) a_1 (C) a_n (D) does not exist

DF0094

Paragraph for Question 4 to 6

Consider the function defined implicitly $y^2 + y - x = 2$ on various intervals on the real line. If $y \in \left(-\infty, -\frac{1}{2}\right)$, then equation implicitly defines a unique real valued differentiable function $y = f(x)$.

On the basis of above informations, answer the following questions.

4. The value of $f''(4)$ is -

- (A) $\frac{2}{125}$ (B) $\frac{125}{2}$ (C) $\frac{1}{5}$ (D) $\frac{1}{125}$

DF0125

5. If $y = g(x)$ is inverse of $y = f(x)$, then $g'(-2)$ is -

- (A) $-\frac{1}{3}$ (B) -3 (C) -2 (D) 0

DF0126

6. If $y = h(x)$ is mirror image of $y = f(x)$ about the line $2y + 1 = 0$, then $f''(x_0) + h''(x_0)$, where $x_0 \in \left(-\frac{9}{4}, \infty\right)$ is -

- (A) $2x_0$ (B) $\frac{2}{2f(x_0)+1}$ (C) $\frac{2}{(2f(x_0)+1)^3}$ (D) 0

DF0127

Paragraph for Question 7 and 8

Let $f : (0, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be a function defined as $f(x) = \tan^{-1}(\ell nx)$ and g be the inverse function of f , then

7. The value of $\frac{|g''(0)|}{2}$ is

DF0128

8. Let $H(x) = \frac{g''(x) - g'(x)\sec^2 x}{e^{\tan x} \cdot \sec^2 x}$, then the value of $H'(0)$ is

DF0129

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) Let $f(x) = x^2 - 4x - 3$, $x > 2$ and let g be the inverse of f .

(1) -1

Find the value of g' at $x = 2$.

(Q) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$

(2) $\frac{1}{6}$

and $g(x) = f^{-1}(x)$, then find $g'(-1)$.

(R) Suppose f^{-1} is the inverse function of a differentiable function f

(3) 1

and let $G(x) = \frac{1}{f^{-1}(x)}$. If $f(3) = 2$ and $f'(3) = \frac{1}{9}$, find $G'(2)$.

(S) If $y = y(x)$ and it follows the relation $e^{xy} + y \cos x = 2$,
then $y'(0) + y''(0)$.

(4) $\frac{1}{14}$

(A) $P \rightarrow 4$; $Q \rightarrow 1$; $R \rightarrow 2$; $S \rightarrow 3$

(B) $P \rightarrow 2$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 4$

(C) $P \rightarrow 2$; $Q \rightarrow 4$; $R \rightarrow 1$; $S \rightarrow 3$

(D) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 1$

DF0130

10. Match the following List-I with List-II

List-I

List-II

(P) Let $f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = x^2 + xf'(2) + f''(3)$,
then $f'(1) - f'(2)$ is equal to

(1) 2

(Q) If $f(x - y)$, $f(x) \cdot f(y)$ and $f(x + y)$ are in A.P., for all x, y
and $f(0) \neq 0$, then $f'(-2) + f'(2)$ is equal to

(2) 1

(R) If $f(x) = x^3 + x^2$, $f'(1) + xf''(2) + f'''(3)$ for all, then
 $f(0) + f(3) + 1$ is equal to

(3) 4

(S) Let $f(x) = x^n$, n being a positive integer,
then value of n for which the equality
 $f'(a + b) = f'(a) + f'(b)$ is valid for all $a, b > 0$ is

(4) 0

(A) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 1$

(B) $P \rightarrow 2$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 1$

(C) $P \rightarrow 3$; $Q \rightarrow 4$; $R \rightarrow 2$; $S \rightarrow 1$

(D) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 3$; $S \rightarrow 4$

DF0131

EXERCISE (O-4)

1. Let $y = 1 + \frac{x_1}{x - x_1} + \frac{x_2 \cdot x}{(x - x_1)(x - x_2)} + \frac{x_3 \cdot x^2}{(x - x_1)(x - x_2)(x - x_3)} + \dots$ upto $(n + 1)$ terms

and if $\frac{x}{y} \frac{dy}{dx} = k \left[\frac{x_1}{x_1 - x} + \frac{x_2}{x_2 - x} + \frac{x_3}{x_3 - x} + \dots + \frac{x_n}{x_n - x} \right]$, then k is

DF0051

2. Let $x = \operatorname{cosec} \theta - \sin \theta$; $y = \operatorname{cosec}^6 \theta - \sin^6 \theta$ if $(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - k(y^2 + 4) = 0$, then k is

DF0052

3. If a curve is represented parametrically by the equations

$$x = \sin \left(t + \frac{7\pi}{12} \right) + \sin \left(t - \frac{\pi}{12} \right) + \sin \left(t + \frac{3\pi}{12} \right),$$

$$y = \cos \left(t + \frac{7\pi}{12} \right) + \cos \left(t - \frac{\pi}{12} \right) + \cos \left(t + \frac{3\pi}{12} \right) \text{ then find the value of } \frac{d}{dt} \left(\frac{x}{y} - \frac{y}{x} \right) \text{ at } t = \frac{\pi}{8}.$$

DF0053

4. Let $\sqrt{1 - x^6} + \sqrt{1 - y^6} = a^3 \cdot (x^3 - y^3)$ and if $\frac{dy}{dx} = \frac{x^\ell}{y^\ell} \sqrt{\frac{1 - y^k}{1 - x^k}}$ (where $x \neq \pm 1$), then $\frac{k}{\ell}$ is

DF0057

5. Let $f(x) = x + \frac{1}{2x + 2x + 2x + \dots} \infty$, then the value of $f(100) \cdot f'(100)$.

DF0059

6. If $y = \tan^{-1} \frac{1}{x^2 + x + 1} + \tan^{-1} \frac{1}{x^2 + 3x + 3} + \tan^{-1} \frac{1}{x^2 + 5x + 7} + \tan^{-1} \frac{1}{x^2 + 7x + 13} + \dots$ to 10 terms. Find $(f'(0) + 1)^{-1}$.

DF0065

7. If $y = \tan^{-1} \frac{u}{\sqrt{1 - u^2}}$ & $x = \sec^{-1} \frac{1}{2u^2 - 1}$, $u \in \left(0, \frac{1}{\sqrt{2}} \right) \cup \left(\frac{1}{\sqrt{2}}, 1 \right)$ then value of $2 \frac{dy}{dx} + 1$ is

DF0066

8. If $y = \cot^{-1} \frac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}}$, find $\frac{dy}{dx}$ if $x \in \left(0, \frac{\pi}{2} \right) \cup \left(\frac{\pi}{2}, \pi \right)$.

DF0067

9. A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions $f(0) = 2$; $f'(0) = -5$ and $f''(0) = 3$.

The function $g(x)$ is defined by $g(x) = e^{ax} + f(x) \forall x \in \mathbb{R}$, where 'a' is any constant. If $g'(0) + g''(0) = 0$, then number of possible value(s) of 'a' is

DF0070

10. If $x = 2\cos t - \cos 2t$ & $y = 2\sin t - \sin 2t$, find the value of (d^2y/dx^2) when $t = (\pi/2)$.

DF0071

EXERCISE (JM)

1. $\frac{d^2x}{dy^2}$ equals :- [AIEEE-2011]

(1) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (2) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$ (3) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (4) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$

DF0098

2. If $y = \sec(\tan^{-1}x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to : [JEE(Main)-2013]

(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$ (3) 1 (4) $\sqrt{2}$

DF0099

3. If g is the inverse of a function f and $f'(x) = \frac{1}{1+x^5}$, then $g'(x)$ is equal to : [JEE(Main)-2014]

(1) $1+x^5$ (2) $5x^4$ (3) $\frac{1}{1+\{g(x)\}^5}$ (4) $1+\{g(x)\}^5$

DF0100

4. If for $x \in \left(0, \frac{1}{4}\right)$, the derivative of $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x} \cdot g(x)$, then $g(x)$ equals :- [JEE(Main)-2017]

(1) $\frac{3}{1+9x^3}$ (2) $\frac{9}{1+9x^3}$ (3) $\frac{3x\sqrt{x}}{1-9x^3}$ (4) $\frac{3x}{1-9x^3}$

DF0101

5. If $x = 3 \tan t$ and $y = 3 \sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is: [JEE(Main)-2019]

(1) $\frac{3}{2\sqrt{2}}$ (2) $\frac{1}{3\sqrt{2}}$ (3) $\frac{1}{6}$ (4) $\frac{1}{6\sqrt{2}}$

DF0102

6. Let $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$ and $g(x) = |f(x)| + f(|x|)$. Then, in the interval $(-2, 2)$, g is :-

[JEE(Main)-2019]

- (1) differentiable at all points (2) not differentiable at two points
(3) not continuous (4) not differentiable at one point

DF0103

7. For $x > 1$, if $(2x)^{2y} = 4e^{2x-2y}$, then $(1 + \log_e 2x)^2 \frac{dy}{dx}$ is equal to : [JEE(Main)-2019]

(1) $\log_e 2x$ (2) $\frac{x \log_e 2x + \log_e 2}{x}$ (3) $x \log_e 2x$ (4) $\frac{x \log_e 2x - \log_e 2}{x}$

DF0104

8. If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at $x = 0$ is equal to : [JEE(Main)-2019]

(1) $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$ (2) $\left(\frac{1}{e}, \frac{1}{e^2}\right)$ (3) $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ (4) $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$

DF0106

9. The derivative of $\tan^{-1}\left(\frac{\sin x - \cos x}{\sin x + \cos x}\right)$, with respect to $\frac{x}{2}$, where $\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ is : [JEE(Main)-2019]

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) 1 (4) 2

DF0107

10. Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$.

Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to :

[JEE(Main)-2020]

(1) $\frac{\sqrt{5}}{2}$ (2) $-\frac{\sqrt{5}}{2}$ (3) $\frac{2}{\sqrt{5}}$ (4) $-\frac{\sqrt{5}}{4}$

DF0132

11. If $y(\alpha) = \sqrt{2\left(\frac{\tan \alpha + \cot \alpha}{1 + \tan^2 \alpha}\right) + \frac{1}{\sin^2 \alpha}}$, $\alpha \in \left(\frac{3\pi}{4}, \pi\right)$, then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is : [JEE(Main)-2020]

(1) 4 (2) $-\frac{1}{4}$ (3) $\frac{4}{3}$ (4) -4

DF0133

12. If $y = \sum_{k=1}^6 k \cos^{-1}\left\{\frac{3}{5}\cos kx - \frac{4}{5}\sin kx\right\}$, then $\frac{dy}{dx}$ at $x = 0$ is _____. [JEE(Main)-2020]

DF0134

13. Consider the function $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)} & ; x \neq 2 \\ 7 & ; x = 2 \end{cases}$

Where $P(x)$ is a polynomial such that $P''(x)$ is always a constant and $P(3) = 9$. If $f(x)$ is continuous at $x = 2$, then $P(5)$ is equal to _____. [JEE(Main)-2021]

DF0135

14. Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$, $0 < x < 1$. Then : [JEE(Main)-2021]

(1) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$ (2) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$
(3) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$ (4) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$

DF0136

15. If $y(x) = \cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right)$, $x \in \left(\frac{\pi}{2}, \pi\right)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{6}$ is : [JEE(Main)-2021]

(1) $-\frac{1}{2}$ (2) -1 (3) $\frac{1}{2}$ (4) 0

DF0137

EXERCISE (JA)

1. Let $g(x) = \ln f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = x f(x)$. Then for $N = 1, 2, 3, \dots$,

$$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$$

[JEE 2008, 3]

$$(A) -4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$$

$$(B) 4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$$

$$(C) -4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$$

$$(D) 4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$$

DF0108

2. Let f and g be real valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous, $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$, and $f(x) = g(x) \sin x$.

Statement-1 : $\lim_{x \rightarrow 0} [g(x) \cot x - g(0) \operatorname{cosec} x] = f''(0)$

and

Statement-2 : $f'(0) = g(0)$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation of statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

[JEE 2008, 3]

DF0109

3. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is

[JEE 2009, 4]

DF0110

4. Let $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan \theta)}(f(\theta))$ is

[JEE 2011, 4]

DF0111

5. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point $(1, 3)$ is

[JEE(Advanced)-2014, 3]

DF0112

6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then-

[JEE(Advanced)-2016, 4(-2)]

$$(A) g'(2) = \frac{1}{15}$$

$$(B) h'(1) = 666$$

$$(C) h(0) = 16$$

$$(D) h(g(3)) = 36$$

DF0113

ANSWER KEY

Do yourself -1 :

1. (a) $\frac{1}{x}$ (b) $-\frac{1}{x^2}$ (c) $\frac{\sec^2 \sqrt{x}}{2\sqrt{x}}$ (d) $-\sin(\sin x) \cdot \cos x$
 (e) $\frac{-1}{\sqrt{1-x^2}}, (-1 < x < 1)$ (f) $\frac{1}{|x|\sqrt{x^2-1}}, |x| > 1$

Do yourself -2 :

1. (a) $3x^2 + 12x + 11$ (b) $5e^{5x} \tan(x^2 + 2) + 2xe^{5x} \sec^2(x^2 + 2)$
 2. A 3. A 4. B 5. B 6. A 7. C 8. C
 9. C 10. C 11. A 12. B 13. B 14. D
 15. $\frac{\sin x + \cos x + x(\sin x - \cos x)}{1 + \sin 2x}$ 16. $\tan^4 x$ 17. $2x \frac{\sin x}{\cos^3 x}$
 18. $-\frac{a}{3} \sin \frac{x}{3}$ 19. $\frac{1}{2 \cos^2 \frac{x+1}{2}}$ 20. $\frac{1}{\sqrt{1+2 \tan x \cdot \cos^2 x}}$
 21. $-12 \cos^2 4x \sin 4x$ 22. $\frac{1}{4 \sqrt{\tan \frac{x}{2} \cdot \cos^2 \frac{x}{2}}}$ 23. $\frac{x \cos \sqrt{1+x^2}}{\sqrt{1+x^2}}$
 24. $-\frac{2x}{3 \sin^2 \sqrt[3]{1+x^2} \cdot \sqrt[3]{(1+x^2)^2}}$ 25. $4(1 + \sin^2 x)^3 \sin 2x$
 26. $\frac{x^2 - 1}{2x^2 \cos^2 \left(x + \frac{1}{x}\right) \sqrt{1 + \tan \left(x + \frac{1}{x}\right)}}$ 27. $\frac{\sin 2 \left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)}{\sqrt{x}(1+\sqrt{x})^2}$ 28. $-3 \sin 3x \sin(2 \cos 3x)$
 29. $\arcsin x + \frac{x}{\sqrt{1-x^2}}$ 30. $\frac{\pi}{2(\arccos x)^2 \sqrt{1-x^2}}$ 31. $\frac{2 \arcsin x}{\sqrt{1-x^2}}$
 32. $-\frac{1}{(\arcsin x)^2 \sqrt{1-x^2}}$ 33. $\sin x \cdot \arctan x + x \cos x \cdot \arctan x + \frac{x \sin x}{1+x^2}$
 34. 0 35. $-\frac{2x^2}{(1+x^2)^2}$ 36. $\frac{2x}{1+x^4}$
 37. $\frac{x+1}{8 \sqrt[4]{(\arcsin \sqrt{x^2+2x})^3 \sqrt{(1-2x-x^2)(x^2+2x)}}$ 38. $\frac{\sqrt{a^2-b^2}}{a+b \cos x}$
 39. $2x \log_3 x + \frac{x}{\ln 3}$ 40. $\frac{1}{2x \sqrt{\ln x}}$
 41. $\sin x \ln x + x \cos x \ln x + \sin x$ 42. $\frac{1-n \ln x}{x^{n+1}}$ 43. $\frac{-2}{x(1+\ln x)^2}$
 44. $\frac{2x-4}{x^2-4x}$ 45. $\frac{2}{\sin 2x}$

46. $\frac{1}{x \log_5 x \log_3 (\log_5 x) \ln 2 \ln 3 \ln 5}$ 47. $\frac{x}{\arctan \sqrt{1+x^2} (2+x^2) \sqrt{1+x^2}}$ 48. $\frac{\cot \frac{x+3}{4}}{12 \sqrt[3]{\ln^2 \sin \frac{x+3}{4}}}$
49. $10^x (1+x \ln 10)$ 50. $\frac{2^x (\ln 2 - 1) + 3x^2 - x^3}{e^x}$ 51. $-\frac{\sin x + \cos x}{e^x}$
52. $\frac{(\ln x - 1) \ln 2}{\ln^2 x} 2^{\frac{x}{\ln x}}$ 53. $-\frac{2 \cdot 10^x \ln 10}{(1+10^x)^2}$ 54. $3^{\sin x} \cos x \cdot \ln 3$
55. $-2ab^2 x e^{-b^2 x^2}$ 56. $A e^{-k^2 x} [\omega \cos(\omega x + \alpha) - k^2 \sin(\omega x + \alpha)]$

Do yourself -3 :

1. $x^x (\ln x + 1)$ 2. $y(1+2x+3x^2+4x^3)$ 3. 5050.00 4. A 5. C
6. A 7. B 8. B 9. $x^{x^x} \cdot x^x \left(\ln^2 x + \ln x + \frac{1}{x} \right)$
10. $(\sin x)^{\cos x} \left(\frac{\cos^2 x}{\sin x} - \sin x \ln \sin x \right)$ 11. $(\ln x)^x \left(\frac{1}{\ln x} + \ln \ln x \right)$
12. $2x^{\ln x - 1} \ln x$ 13. $\frac{57x^2 - 302x + 361}{20(x-2)(x-3)} \cdot \frac{(x+1)^2 \sqrt[4]{x-2}}{\sqrt[5]{(x-3)^2}}$
14. $\frac{1}{2} \sqrt{x \sin x \sqrt{1-e^x}} \left(\frac{1}{x} + \cot x - \frac{1}{2} \cdot \frac{e^x}{1-e^x} \right)$ 15. $x^{\frac{1}{x}-2} (1 - \ln x)$ 16. $x^{\sqrt{x}-\frac{1}{2}} (2 + \ln x)$
17. $(x^2+1)^{\sin x} \left[\frac{2x \sin x}{x^2+1} + \cos x \ln(x^2+1) \right]$ 18. $\frac{x^4+6x^2+1}{3x(1-x^4)} \sqrt[3]{\frac{x(x^2+1)}{(x^2-1)^2}}$

Do yourself-4 :

1. -1 2. $\frac{1}{t}$ 3. $2(x^{\ln x})(\ln x)$ 4. A 5. C 6. C
7. D 8. C

Do yourself -5 :

1. $\frac{\cos(x-y)-1}{\cos(x-y)+1}$ 2. $y' = -\left(\frac{2x+e^y}{xe^y+1} \right), -1$ 3. D 4. C 5. C
6. A 7. B 8. C 9. D 10. D 11. A 12. 9.00
13. 2011.00 14. A 15. $\frac{y}{x} \cdot \frac{x \ln x + x \ln x \cdot \ln y + 1}{\ln x (1-x-y \ln a)}$

Do yourself : 6

1. (a) -6 (b) 3 (c) -6 2. A 3. C 4. A 5. C
6. A 7. C 8. A 9. D 10. D 11. D 12. C 13. A

Do yourself -7 :

1. D 2. 1 3. 0 4. (B) 5. $g'(9) = \frac{1}{35}, g''(9) = \frac{-118}{(35)^3}$ 6. 8

1. $y'' = 4y + 2xy'$ 2. $\pi + 4$ 3. Do yourself : 8
4. D

Do yourself : 9

1. $e(\sin 1 + \cos 1) - 1$ 2. 9 3. A 4. D 5. D 6. B

Do yourself : 10

1. (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ 3. D 4. A 5. 1000 6. 11
7. f is cont. but not derivable at $x = 0$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	B	C	D	A	A	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	D	A	A	D	A	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	A	D	C	3	4	4	2	A	2

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	C,D	A,B,C	A,C	A,B	A,C,D	A	A,B,C	B,D	A,B,C,D
Que.	11	12	13	14	15					
Ans.	A,B	A,C	B,C,D	A,D	A,B,C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	C	A	B	D	0.50	2.00	C	C

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	36	8	3	100	101.00	0.00	0.50, -0.50	2.00	-1.50

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	4	2	4	4	4	1	4	2
Que.	11	12	13	14	15					
Ans.	1	91	39	3	1					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6				
Ans.	A	A	2	1	8	B,C				

Important Notes

